Airline customer satisfaction: a bayesian prediction model

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Abstract

In this project we define and fit a Bayesian prediction model for the satisfaction level of customers of an undisclosed airline, based on a set of flight characteristics. Model parameters are derived using MCMC methods via the JAGS tool. We compare a logistic regression under weakly informative priors against a horseshoe shrinkage prior over 22 predictors, assess both with WAIC and PSIS-LOO, and validate the selected model on a held-out set of 128 274 *customers.*

Contents

1. Problem description and data exploration

1.1 Problem description

The goal of this project is to predict the customer satisfaction level of an unspecified airline, given a set of covariates X that describe several characteristics of the flight, such as service-quality items, type of customer, departure delay and so on, and customer's opinion on the flight, like the rating given to some service quality items. Given the high degree of uncertainty inherent in personal opinions, we chose to adopt a Bayesian logistic regression model[1]. The Bayesian coefficients are estimated via Markov Chain Monte Carlo (MCMC) methods.

1.2 Dataset description

The dataset [2] contains 129,881 customer records, described by 22 columns, organised as follows:

- **Target variable**: represents the satisfaction level of customer i, denoted y_i, which can be either *dissatisfied* or *satisfied* and is encoded respectively as $\{0,1\}$.
- **Categorical variables (3)**: indicate the category of the customer and of the flight to which the record refers. For these variables, one-hot encoding is used.
  Variable list: `Customer type, Type of Travel, Class`.
- **Discrete variables (15)**: mostly composed of quality-of-service variables.
  Variable list (service-quality items excluded): `Age`.
  - **Service-quality variables (14)**: indicate the various levels of service available throughout the whole journey, from booking to arrival. These variables contain several zeros, which may indicate that the corresponding service was not available during the flight.
    Variable list: `Seat comfort, Departure/Arrival time convenience, Food and drink, Gate location, Inflight wifi service, Inflight entertainment, Online support, Ease of Online booking, On-board service, Leg room service, Baggage handling, Check-in service, Cleanliness, Online boarding`.
- **Flight properties (3)**: report the spatial and temporal properties of the flight. They are treated as continuous variables.
  Variable list: `Flight distance, Departure delay, Arrival delay`.
- **Modified variables**:
  - owing to the extremely high correlation between `Departure delay` and `Arrival delay`, we decided to replace the latter with the difference between the two delays, so as to retain the information while reducing the correlation.
  - given the high skewness of variables `Departure Delay` and `Mid-flight delay`, we decided to correct it trough logarithm transormation. `Flight Distance` was tested but kept on its original scale as transforming it worsened rather than improving the prediction.

Further details on the procedures are given in Appendix A.

1.3 Dataset exploration

We begin our exploration by examining the balance of the target variable for the prediction model. As shown in Fig. 1, the two classes are fairly balanced, so the use of models more complex than a Bernoulli probability distribution is not justified and will not be pursued.

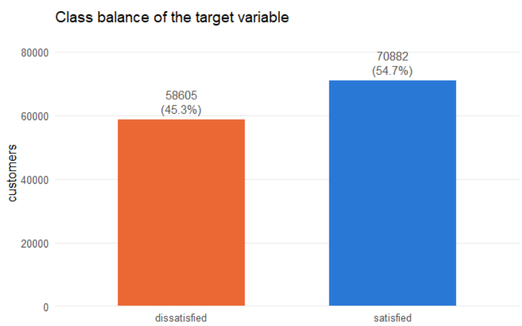
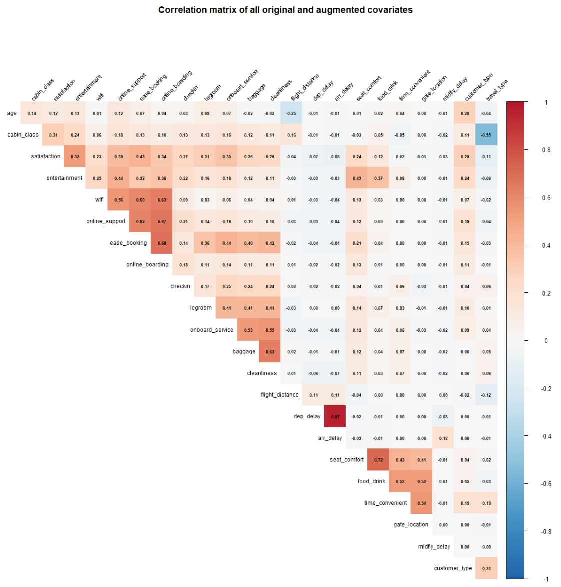
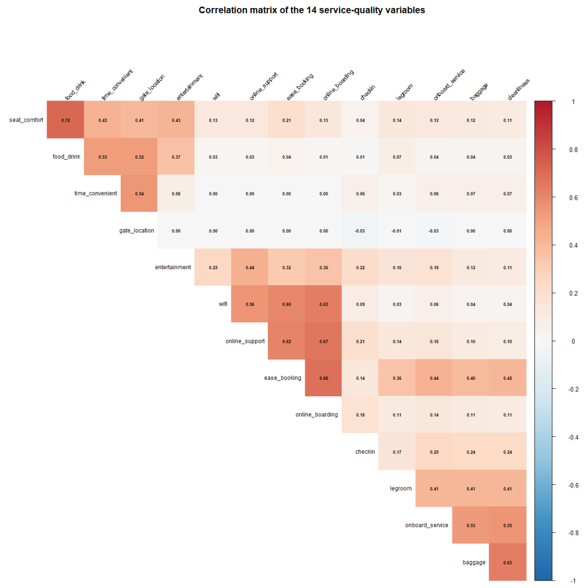


Figure 1. Target variable class balance



(a) Correlation among all variables, both original and augmented



(b) Correlation among service-quality items

To obtain a first indication of the relationship between the response variable and the remaining covariates, we examine their correlation, as shown in Fig. 2a. Focusing on the service-quality variables (Fig. 2b), we can distinguish three subgroups of features. The correlation is computed using R's `corr()` function, which is based on Pearson's correlation coefficient

$$r_{X,Y} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

Tab. 1 reports the class balance of the remaining categorical variables. Given the imbalance among their levels, we decided to stratify the train–test split of the dataset.

Table 1. Class balance of the categorical covariates

Variable	Level	Proportion
Satisfaction	Dissatisfied	0.453
	Satisfied	0.547
Customer type	Disloyal customer	0.183
	Loyal customer	0.817
Travel type	Business travel	0.691
	Personal travel	0.309
Cabin class	Eco	0.449
	Eco Plus	0.072
	Business	0.479

2. Logit model

In this chapter we present the results obtained by fitting a simple Bayesian logistic regression on a reduced set of covariates. The aim is twofold: to establish a first, fully interpretable baseline for the satisfaction prediction problem introduced in Section 1, and to settle a few modelling choices (prior specification, encoding of the cabin class, MCMC setup) that will be reused by the richer shrinkage model of Section 3.

2.1 Covariates and model specification

Following the suggestions of the brief, we fit the model on a compact subset of covariates, using the transformations motivated in the data-exploration phase (Section 1.3): `age`, `seat_comfort`, `flight_distance`, `cabin_class`, the log-transformed departure delay $\log(1+\texttt{dep\_delay})$ and the signed-log mid-flight delay, together with the two binary descriptors `travel_type` and `customer_type`. All continuous covariates are standardised (zero mean, unit variance) so that coefficients are expressed per standard deviation and the common prior variance is equally informative for every covariate. For the binary covariates the reference level is chosen as the majority group (*Business travel* and *Loyal customer*), so that each coefficient measures the effect of the minority condition relative to the typical customer.

Let $y_i \in \{0,1\}$ be the satisfaction outcome of customer i (encoded as in Section 1) and x_i the corresponding row of the standardised design matrix. We assume a Bernoulli likelihood with a logit link:

$$y_i \mid p_i \sim \text{Bernoulli}(p_i), \qquad \text{logit}(p_i) = \beta_0 + x_i^\top \beta + \beta_{\text{mid}}\,\texttt{midflight\_delay}_i. \tag{1}$$

The mid-flight delay is deliberately kept outside the main design matrix and given its own coefficient β_{mid}, paired with an explicit per-row sampling statement $\texttt{midflight\_delay}_i \sim \mathcal{N}(0,1)$. This treats the (standardised) mid-flight delay as a stochastic node rather than a fixed regressor, which is convenient for handling it as an augmented quantity and keeps its contribution cleanly separated from the other effects.

Prior specification. Since we have no strong prior belief on the magnitude or the sign of the coefficients, and because standardisation already puts every covariate on a comparable scale, we adopt independent, weakly informative Gaussian priors on all the parameters:

$$\beta_0 \sim \mathcal{N}(0,100), \qquad \beta_j \sim \mathcal{N}(0,100)\ \ (j=1,\dots,P), \qquad \beta_{\text{mid}} \sim \mathcal{N}(0,100). \tag{2}$$

Here the second argument denotes the prior *variance* $\sigma^2 = 100$ (equivalently a precision $\tau = 1/\sigma^2 = 0.01$ in the JAGS parametrisation). On the logit scale a standard deviation of 10 is very diffuse: it places essentially no informative constraint on plausible coefficient values while still being proper, so the posterior is driven by the likelihood, and the resulting estimates remain directly comparable to a maximum-likelihood fit. This choice reflects our lack of prior information and keeps the baseline model as neutral as possible.

2.2 Encoding of the cabin class

Before committing to the model we address a small preliminary question: is it preferable to encode `cabin_class` as a pair of dummy variables (*Eco Plus* and *Business*, with *Eco* as reference) or as a single centred ordinal score $(1/2/3)$? To decide, we fit the model (1) under both encodings and compare them with the Widely Applicable Information Criterion (WAIC), computed from the pointwise log-likelihood of the MCMC draws.

The dummy encoding attains the lower WAIC (1373.2 against 1379.6 for the ordinal one), i.e. a slightly better expected predictive fit. The difference in expected log predictive density is $\Delta\text{elpd} = -3.2$ with a standard error of 2.7, so the preference, although real, is smaller than two standard errors and therefore modest. Because `cabin_class` is genuinely *nominal* rather than an equally-spaced ordinal scale, and because the dummy specification is both more flexible and lets us report a separate, directly interpretable odds ratio for each class, we adopt the dummy encoding for the rest of the analysis.

2.3 MCMC setup and convergence

Posterior inference is carried out via MCMC in JAGS. We run 3 parallel chains, with an adaptation phase of 1500 iterations, a burn-in of 1000 iterations, 3000 monitored iterations per chain and a thinning factor of 7 to reduce the residual autocorrelation between consecutive draws. Table 11 summarises these choices.

Table 2. MCMC settings used for the logit model

Setting	Value
Chains	3
Adaptation	1500
Burn-in	1000
Iterations (kept)	3000
Thinning	7

Convergence was satisfactory for all parameters. The Gelman–Rubin potential scale reduction factors are essentially 1 (point estimates 1.00–1.01, upper confidence limits up to 1.02, multivariate PSRF = 1.01), the trace plots show well-mixed, stationary chains overlapping around a common region, and the autocorrelation decays quickly to zero within a few lags after thinning. The effective sample sizes range from about 590 (for the intercept and the personal-travel coefficient) to roughly 1350, which is more than adequate for stable posterior summaries. A complete convergence assessment (trace plots, autocorrelation functions and the full diagnostic tables) is deferred to Appendix C.

2.4 Posterior results

Figures for the posterior densities of the coefficients and the odds-ratio forest plot are collected below; the corresponding numerical summaries are reported in Table 3. For each coefficient we report the posterior mean, the 95% credible interval on the log-odds scale (β) and the same quantities transformed to the odds-ratio scale ($\mathrm{OR} = e^{\beta}$), which is the natural scale for interpretation.

Table 3. Posterior summaries of the logit coefficients and their odds ratios, sorted by decreasing OR. CIs are 95% credible intervals.

Term	$\bar{\beta}$	$\beta_{2.5\%}$	$\beta_{97.5\%}$	OR	$\mathrm{OR}_{2.5\%}$	$\mathrm{OR}_{97.5\%}$
Class: Business (vs Eco)	1.450	1.120	1.804	4.265	3.065	6.075
Seat comfort	0.632	0.485	0.768	1.881	1.625	2.156
Class: Eco Plus (vs Eco)	0.050	−0.441	0.524	1.052	0.643	1.688
Age	−0.026	−0.164	0.127	0.975	0.848	1.135
Arrival delay residual (signed-log)	−0.041	−0.187	0.111	0.960	0.830	1.118
Flight distance	−0.052	−0.185	0.090	0.949	0.831	1.094
Intercept	−0.105	−0.400	0.180	0.901	0.670	1.198
Personal Travel (vs Business travel)	−0.168	−0.498	0.166	0.846	0.608	1.181
$\log(1+\text{Departure Delay})$	−0.230	−0.371	−0.078	0.795	0.690	0.925
disloyal Customer (vs Loyal)	−1.825	−2.253	−1.431	0.161	0.105	0.239

2.5 Interpretation

Four effects clearly separate from the "no-effect" line (OR = 1). By far the largest is the cabin class: travelling in *Business* rather than *Eco* multiplies the odds of being satisfied by roughly 4.3 (95% CI $[3.07, 6.08]$), even after adjusting for all the other covariates, confirming the marginal contrast already seen in the data-exploration phase. *Seat comfort* is the strongest continuous driver (OR ≈ 1.88 per standard deviation, $[1.63, 2.16]$): more comfortable seating is strongly associated with satisfaction. In the opposite direction, a *disloyal customer* has markedly lower odds of being satisfied (OR ≈ 0.16, $[0.11, 0.24]$), and a longer departure delay reduces them too (OR ≈ 0.80 per unit of $\log(1+\text{delay})$, $[0.69, 0.93]$).

The remaining covariates have credible intervals that comfortably include 1 and cannot be called confident drivers on their own: *Eco Plus* is statistically indistinguishable from *Eco*, while *age*, *flight distance* and the *mid-flight delay residual* all show weak, uncertain effects.

The most instructive case is the *type of travel*. In the exploratory analysis personal-travel customers appeared markedly less satisfied, an almost certain marginal association; here, once cabin class and the other covariates are held fixed, the credible interval for *Personal Travel* (0.61 to 1.18) comes close to, but no longer excludes, 1. This is exactly the confounding anticipated in Section 1.3: business travellers are disproportionately booked in Business class, so once class enters the model directly it absorbs most of the apparent travel-type effect. Both the borderline effects (flight distance, departure delay) and this partially-explained travel-type signal motivate the richer specification of Section 3, where the full battery of 14 service-quality ratings is added under a shrinkage prior to clarify which of these effects survive alongside the stronger, correlated service signals.

3. Horseshoe shrinkage model

3.1 Motivation

The logit model of Section 2 was fit on the small set of covariates suggested by the brief, taking their usefulness for granted. That is a reasonable baseline, but it leaves open the central question of the analysis: among *all* the available predictors, in particular the fourteen service-quality ratings that were set aside earlier, which ones actually carry signal for customer satisfaction, and which are redundant? Rather than answering this by hand, adding and removing covariates and guessing at their relevance, we prefer a principled, automatic mechanism. We therefore fit a Bayesian logistic regression on the *full* set of covariates under a **horseshoe** shrinkage prior [1], which is designed to pull the coefficients of noise predictors strongly towards zero while leaving genuine signals almost untouched. This lets the data, rather than our prior guesses, single out the meaningful covariates.

3.2 Model specification

We include all 21 standardised predictors: the 14 service-quality items plus `age`, `flight_distance`, the two cabin-class dummies (*Eco Plus*, *Business*), $\log(1+\texttt{dep\_delay})$, *Personal Travel* and *disloyal Customer* - and, as in Section 2, the mid-flight delay is kept outside the design matrix with its own coefficient β_{arr}. The likelihood is again Bernoulli with a logit link:

$$y_i \mid p_i \sim \text{Bernoulli}(p_i), \qquad \text{logit}(p_i) = \beta_0 + x_i^\top \beta + \beta_{\text{arr}}\, \texttt{arr\_delay}_i. \tag{3}$$

The horseshoe prior replaces the common vague Gaussian of Section 2 with a scale-mixture of normals: each coefficient gets its own *local* scale λ_j, and all local scales share a single *global* scale τ,

$$\beta_j \mid \lambda_j, \tau \sim \mathcal{N}(0, \lambda_j^2 \tau^2), \qquad \lambda_j \sim \mathcal{C}^+(0,1), \qquad \tau \sim \mathcal{C}^+(0,1), \tag{4}$$

where $\mathcal{C}^+(0,1)$ denotes the standard half-Cauchy distribution. The global scale τ pulls *all* coefficients towards zero, enforcing overall sparsity, while the heavy Cauchy tails of the local scales λ_j let individual coefficients escape that shrinkage when the data demand it - this is precisely the "a few large, most near zero" behaviour that makes the prior suited to feature selection. The arrival-delay coefficient β_{arr} is given its own local scale λ_{arr} under the *same* global τ, so that it competes for signal on an equal footing with the other 21 predictors. Following the JAGS notes, each half-Cauchy is implemented as a Student-t with one degree of freedom truncated to the positive reals, `dt(0, 1, 1)T(0, )`. The intercept keeps a diffuse Gaussian prior, $\beta_0 \sim \mathcal{N}(0, 100)$ (i.e. `dnorm(0, 0.01)`), and is deliberately left out of the shrinkage so that the baseline log-odds is not penalised.

A convenient by-product of the mixture form is the *shrinkage weight*

$$\kappa_j = \frac{1}{1+\lambda_j^2}, \qquad \kappa_j \in (0,1), \tag{5}$$

which measures how much coefficient j is pulled towards zero: κ_j near 0 marks a preserved *signal* (essentially unshrunk), whereas κ_j near 1 marks a coefficient shrunk towards *noise*. We use the posterior distribution of the κ_j as our covariate-relevance summary.

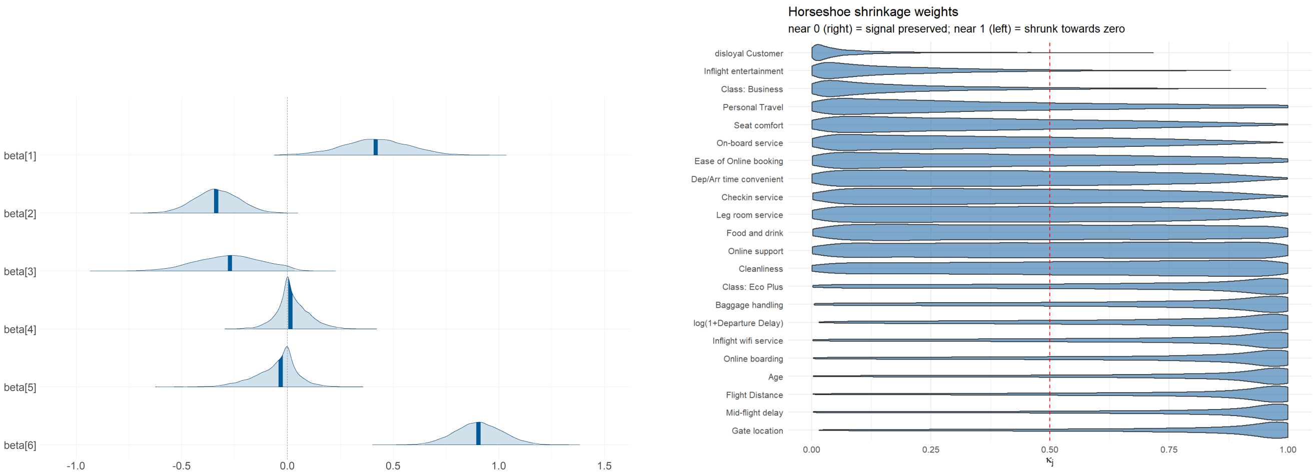
3.3 MCMC setup and convergence

Because of the heavy-tailed half-Cauchy hyperpriors, the horseshoe posterior mixes more slowly than the vague-prior model of Section 2, and it is computationally heavier (22 coefficients, 22 local scales and one global scale, against the 9 parameters of Section 2). We therefore budget a longer sampling run and thinning from the outset; the settings, listed in Appendix C, are deliberately modest given the single-core environment and could be extended on a larger machine.

Convergence was assessed separately for the two blocks of parameters, the regression coefficients $\{\beta_j, \beta_0, \beta_{\text{arr}}\}$ and the scale parameters $\{\lambda_j, \lambda_{\text{arr}}, \tau\}$, since the latter are the harder to sample. For each block we inspected the Gelman–Rubin potential scale reduction factors, the effective sample sizes, and the trace and autocorrelation plots. The coefficients mix well and show scale-reduction factors close to 1; the scale parameters, and τ in particular, mix more slowly and show higher autocorrelation, as expected for a horseshoe, but reach effective sample sizes adequate for the summaries we report. The complete diagnostic material - representative trace plots for each block, all autocorrelation plots for both the β and the λ parameters, together with the full Gelman–Rubin and effective-sample-size tables - is collected in Appendix C.

3.4 Posterior results

Figure 3a shows the marginal posterior densities of the coefficients: most concentrate tightly around zero, with only a handful clearly displaced, which is the visual signature of the shrinkage at work. The relevance ranking is made explicit in Figure 3b, which plots the posterior distribution of the shrinkage weights κ_j of Eq. (5): predictors on the "signal" side (κ_j small) are the ones the model retains, while those pushed to $\kappa_j \to 1$ are treated as redundant. Figure 4 reports the same information on the coefficient scale, as a forest plot of the posterior means and 95% credible intervals of the β_j, colouring in the covariates identified as signal ($\kappa_j < 0.5$).



(a) Marginal posterior densities of the coefficients under the horseshoe prior (98% intervals). Most coefficients are shrunk tightly around zero; only a few escape.

(b) Posterior distribution of the horseshoe shrinkage weights κ_j for each predictor. Near 0 (right) the signal is preserved; near 1 (left) the coefficient is shrunk towards zero. The dashed line marks $\kappa_j = 0.5$.

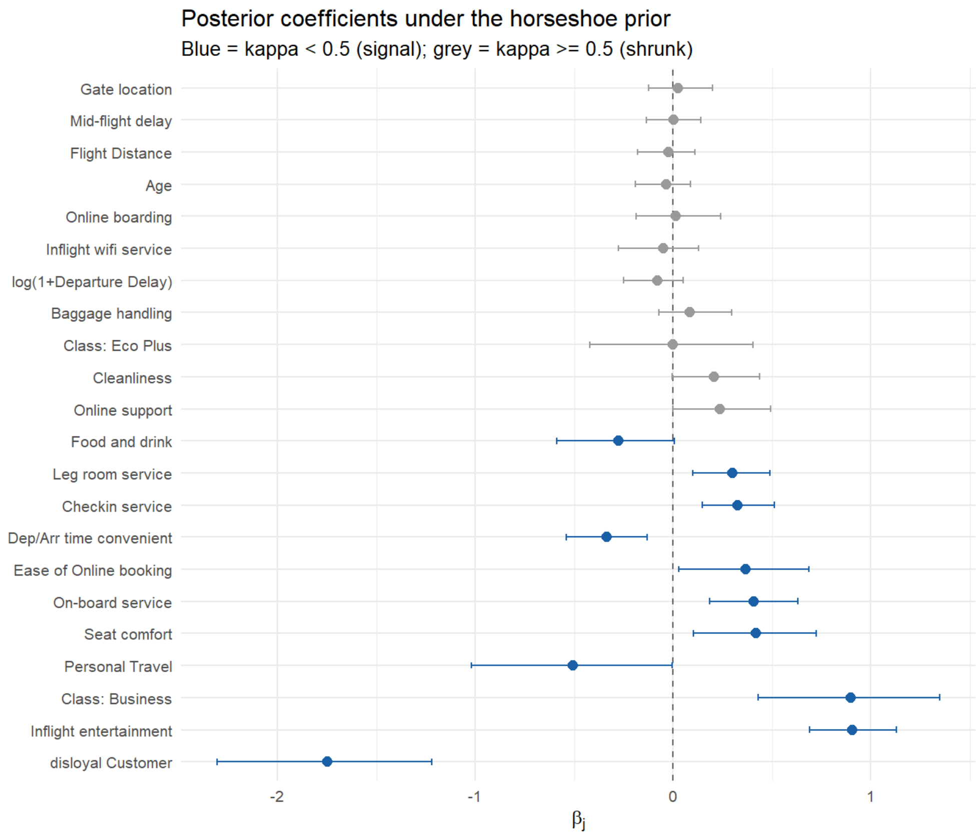


Figure 4. Posterior coefficients β_j under the horseshoe prior (means and 95% credible intervals). Blue marks the covariates identified as signal ($\kappa_j < 0.5$), grey the shrunk ones.

3.5 Conclusions

The horseshoe cleanly separates the informative predictors from the redundant ones, and several of its findings sharpen - and in one case reverse - the picture from Sections 1.3 and 2.

The booking / connectivity cluster visibly competes. Among the four mutually correlated "booking / connectivity" items identified in the exploratory analysis (Section 1.3), the prior does not treat them symmetrically: *Ease of Online booking* and *Online support* are preserved noticeably more than *Online boarding* and *Inflight wifi service* - the latter being the single most shrunk item in the whole model. This is exactly the mechanism the horseshoe was chosen for: when several correlated items carry overlapping information, the prior lets two of them represent the signal and treats the other two as largely redundant, instead of forcing us to pick one by hand.

Business class shrinks once service quality is in the model. In Section 2, *Business* class alone carried a large, confident odds ratio (about 4). Here, with all 14 service ratings included, its coefficient shrinks markedly and its credible interval crosses zero. The natural reading is *mediation*, not disappearance: Business passengers are more satisfied largely *because* they enjoy better seat comfort, entertainment and service, so in Section 2 the *Class* dummy was partly standing in for those ratings simply because they were not yet in the model.

Personal Travel gets stronger, not weaker. Section 2 found the type-of-travel effect nearly explained away by *Class*. Under the horseshoe, once the service ratings control for the *quality* of the trip experience, *Personal Travel* re-emerges as one of the strongest and most confidently negative effects in the model, with a credible interval lying entirely below zero. The two findings are not contradictory: they show that the earlier attenuation was specifically about *Class* soaking up part of the travel-type signal, not about travel type being unimportant.

A cautionary sign on time convenience. More tentatively, *Departure/Arrival time convenient* has a *negative* posterior mean with a credible interval excluding zero - higher ratings associated with *lower* satisfaction, which is counter-intuitive. Recall from Section 1.3 that this item has one of the largest shares of structural zeros and sits in its own correlated cluster with *Seat comfort*, *Food and drink* and *Gate location*; its zeros plausibly encode "not applicable" rather than genuine dissatisfaction. We therefore read this coefficient as a candidate symptom of that ambiguity rather than a reliable substantive finding, and revisit it directly in the structural-zero robustness check of Section 4.3.

Loyalty dominates throughout. Finally, *disloyal Customer* remains, by a wide margin, the single strongest and most confidently preserved effect in the entire model - consistent across the exploratory analysis and both models of Sections 2 and 3, regardless of how many other covariates are present.

4. Prior Sensitivity and Structural Robustness

This section stress-tests the conclusions of Sections 2 and 3 against four choices made along the way: the prior variance of the vague model, the global scale of the horseshoe, the treatment of the ambiguous structural-zero rows, and the size of the training set. Each check requires a full JAGS refit, so they are run by separate scripts (`04a`–`04d`) whose stored output is read back here.

4.1 Logit model prior sensitivity

We refit the model of Section 2 four times, varying only the prior variance on every β_j: $\sigma_\beta^2 \in \{1, 10, 100, 10000\}$. Figure 5 overlays the resulting posterior means and 95% credible intervals, and Table 4 gives the range each posterior mean covers across the four fits.

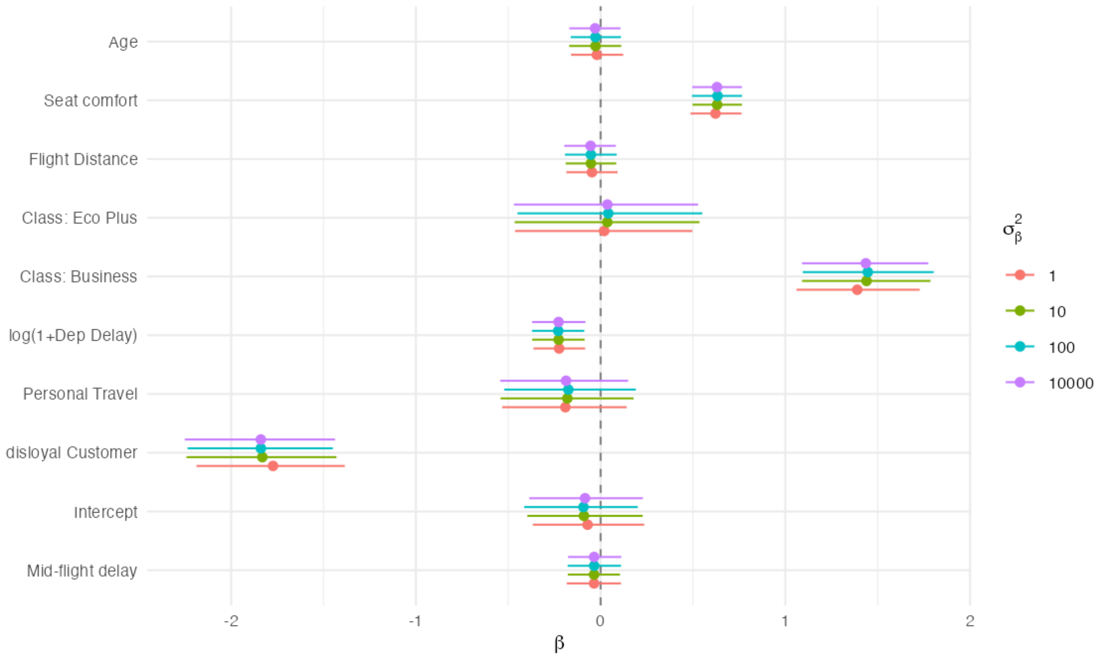


Figure 5. Posterior coefficients of the vague-prior logit model across four prior variances.

Table 4. Range of posterior means across the four prior variances.

Term	Range of posterior mean
disloyal Customer	0.066
Class: Business	0.057
Intercept	0.022
Class: Eco Plus	0.022
Personal Travel	0.016
Seat comfort	0.011
Age	0.009
Flight Distance	0.008
log(1+Departure Delay)	0.005
Mid-flight delay (signed-log)	0.001

Across four orders of magnitude of prior variance the largest movement is 0.066 on the log-odds scale, for a coefficient whose posterior mean is itself of order 1–2. At $n \approx 1200$ the likelihood dominates every prior we tested, so the vague-prior results are not an artefact of the variance we happened to pick.

4.2 Horseshoe prior sensitivity

The horseshoe's behaviour is governed by the scale s of the half-Cauchy prior on the global scale, $\tau \sim \mathscr{C}^+(0, s)$: small s pulls every coefficient towards zero, large s relaxes the shrinkage. It is the one hyperparameter whose value could plausibly change *which* covariates the model keeps, so it is the one worth testing. We refit the Section 3 model at $s \in \{0.1, 1, 10\}$, holding the local scales λ_j at $\mathscr{C}^+(0,1)$, and recompute the shrinkage weights $\kappa_j = 1/(1+\lambda_j^2)$.

Across this hundred-fold range exactly one of the 22 predictors changes its signal/noise verdict, and no posterior mean moves by more than 0.018; the per-predictor weights are tabulated in Appendix C. The strong effects of Section 3 remain

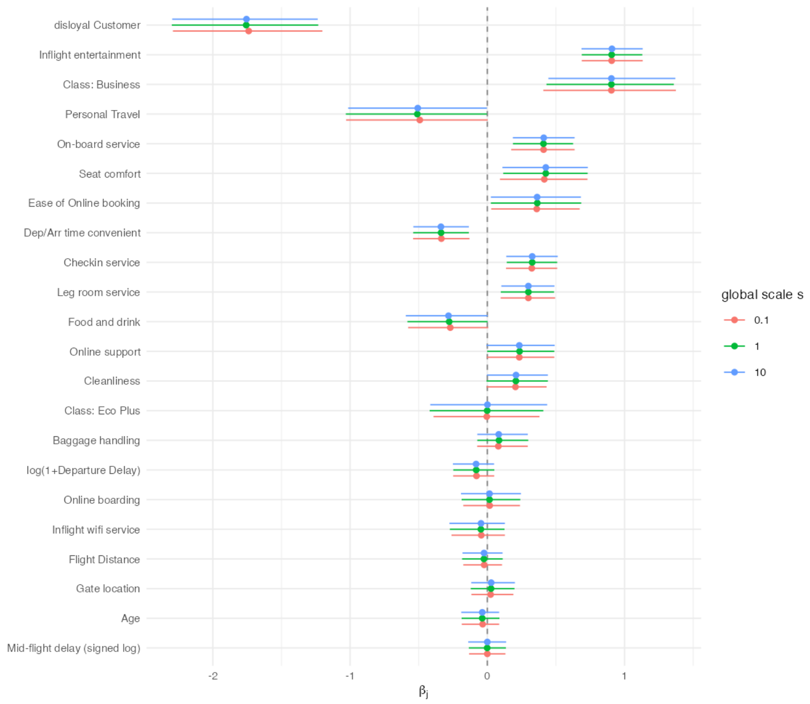


Figure 6. Horseshoe posterior coefficients across three global scales, ordered by shrinkage weight at the default scale.

strong and confidently non-zero at every scale. What little drift there is sits among coefficients already on the $\kappa = 0.5$ boundary, which is where sensitivity to s is expected and where we drew no firm conclusions in the first place.

4.3 Structural-zero robustness check

Section 3 raised the possibility that the negative coefficient on `Departure/Arrival time convenient` is an artefact of the structural zeros flagged in Section 1.3, that is, of zeros meaning "not applicable" rather than genuine dissatisfaction. Refitting the horseshoe model without those rows drops 110 records and leaves 1103.

Table 5. Horseshoe refit excluding structural-zero rows. Only the predictors kept as signal ($\kappa_j < 0.5$) are shown; the full table is in Appendix C.

Term	β mean	β lo	β hi	κ_j
Disloyal Customer	-2.286	-2.975	-1.599	0.059
Inflight entertainment	$+1.090$	$+0.833$	$+1.361$	0.165
Class: Business	$+0.976$	$+0.387$	$+1.535$	0.199
Seat comfort	$+0.881$	$+0.577$	$+1.194$	0.209
Personal Travel	-0.641	-1.278	-0.008	0.331
On-board service	$+0.521$	$+0.231$	$+0.798$	0.338
Dep/Arr time convenient	-0.513	-0.782	-0.234	0.347
Ease of Online booking	$+0.493$	$+0.090$	$+0.902$	0.370
Checkin service	$+0.378$	$+0.154$	$+0.595$	0.420
Leg room service	$+0.297$	$+0.056$	$+0.529$	0.486

The hypothesis does not survive. The coefficient does not shrink towards zero once the ambiguous rows are removed; it becomes *more* negative, from -0.335 to -0.513, with a slightly wider interval as expected with 110 fewer records. Whatever drives the effect, it is not the structural-zero ambiguity we suspected, and we report it as an open finding rather than force a tidier story onto it.

The other two members of that correlated cluster move in opposite directions. `Seat comfort` strengthens, κ falling from 0.367 to 0.209 and β rising from $+0.419$ to $+0.881$, while `Food and drink` goes from a borderline signal ($\kappa = 0.492$, $\beta = -0.275$) to firmly shrunk ($\kappa = 0.733$, $\beta = +0.020$). The structural zeros therefore do not add uniform noise across the cluster; they affect its members differently.

4.4 Sample-size robustness

The checks above varied the prior and which rows enter the fit; this one varies how many. Every model in this report is estimated on 1213 records, following the project brief, which supplies a 1000-customer subset of the original data for exactly this purpose. Refitting the model of Section 2 on a stratified subsample of 15 003 records leaves the coefficients we rely on unmoved, `Class: Business` by 0.005, `disloyal Customer` by 0.014 and `Seat comfort` by 0.022, and contracts the credible intervals by a median factor of 3.42 against the 3.52 that $\sqrt{n}$ asymptotics predicts. Three weak effects (`Flight Distance`, `Age`, `Mid-flight delay`) acquire intervals excluding zero, which is a limitation of the $n = 1213$ analysis rather than a finding about the airline.

5. Prediction and validation

Everything up to this point has been estimated on the 1203-customer training subsample. This section confronts the model with the 128 274 held-out customers set aside in Section 1, which were never used for fitting, standardisation or variable selection. Following the conclusion of Section 4, the horseshoe model is the one carried forward here; the corresponding comparison against the vague-prior logit and against BAS is reported in Appendix D.

5.1 From posterior draws to predicted probabilities

For a held-out customer with covariate row $\tilde{x}_i$, the Bayesian point prediction is the posterior mean of the success probability, obtained by pushing every retained draw through the inverse link and averaging:

$$\hat{p}_i = \frac{1}{S}\sum_{s=1}^{S} \operatorname{logit}^{-1}\Big(\beta_0^{(s)} + \tilde{x}_i^\top \beta^{(s)} + \beta_{\text{arr}}^{(s)}\, \texttt{arr\_delay}_i\Big), \tag{6}$$

where S is the number of retained posterior draws. Note that this is the average of the transformed draws, not the transform of the averaged draws: $\operatorname{logit}^{-1}$ is non-linear, so $\mathbb{E}[g(\theta)] \neq g(\mathbb{E}[\theta])$, and only the former propagates posterior uncertainty into the prediction. The test covariates are standardised with the *training* means and standard deviations, so no information from the held-out set enters the fitted model.

5.2 Discrimination and probabilistic accuracy

Two complementary summaries are reported. The area under the ROC curve (AUC) measures how well the model *ranks* customers by predicted satisfaction and is invariant to the choice of cutoff. The Brier score,

$$\text{BS} = \frac{1}{n}\sum_{i=1}^{n}\big(\hat{p}_i - y_i\big)^2, \tag{7}$$

is a strictly proper scoring rule: it is minimised only when the stated probabilities match the true ones, so it rewards predictions that are both well ranked and numerically honest. On the held-out set the horseshoe model attains an AUC of 0.900 and a Brier score of 0.126.

5.3 Classification and the choice of cutoff

Turning probabilities into hard classifications requires a cutoff. Table 6 contrasts the conventional 0.5 with the training base rate $\bar{y} = 0.546$, the natural alternative when the two classes are not perfectly balanced.

Table 6. Classification performance on the held-out set at two cutoffs.

Cutoff	Sensitivity	Specificity	Accuracy
$t = 0.5$	0.837	0.812	0.826
$t = \bar{y} = 0.546$	0.815	0.843	0.828

Moving the cutoff to the base rate trades about two points of sensitivity for three points of specificity and leaves overall accuracy essentially unchanged (0.826 against 0.828). That near-indifference is itself informative: the errors are not concentrated on one side, and accuracy alone is a blunt instrument for choosing an operating point. What the right cutoff is depends on what the two mistakes cost, which is the subject of Section 5.6.

5.4 Calibration

Neither AUC nor accuracy asks whether the predicted probabilities *mean* what they claim. Calibration does: among the customers the model calls 70% likely to be satisfied, roughly 70% should in fact be satisfied. For a Bayesian analysis this matters as much as discrimination, since the posterior predictive probability is the object actually reported. Table 7 bins the held-out customers into deciles of predicted probability and compares the mean prediction in each bin against the observed satisfaction rate.

The model is well calibrated in the upper half of the range: from the seventh decile upwards predicted and observed rates agree closely (0.801 against 0.819, 0.896 against 0.937). In the middle it is mildly *over*-confident about dissatisfaction, predicting 0.33 and 0.49 in deciles four and five where the observed rates are 0.26 and 0.43. The lowest two deciles reverse the direction, with observed rates (0.122, 0.125) above the predicted ones (0.035, 0.100).

The practical reading is that the model separates the two classes well at the extremes, where most of its predictions live, and is least trustworthy in the band around $\hat{p} \approx 0.3$–0.6 where the evidence genuinely is ambiguous. Any downstream decision quoting a probability from that middle range deserves more caution than the AUC alone would suggest.

5.5 Posterior predictive check by subgroup

The checks above compare point predictions against outcomes. A posterior predictive check asks a stronger question: if the fitted model is taken seriously as a generative description of the data and entire replicated datasets are simulated from it, do those replications reproduce features of the real data that were never targeted by the fit?

Table 7. Calibration by decile of predicted probability.

Decile	Mean predicted	Observed rate	n
1	0.035	0.122	12 828
2	0.100	0.125	12 827
3	0.195	0.163	12 827
4	0.326	0.260	12 828
5	0.489	0.430	12 827
6	0.658	0.637	12 827
7	0.801	0.819	12 828
8	0.896	0.937	12 827
9	0.946	0.984	12 827
10	0.977	0.997	12 828

For each posterior draw we simulate replicated satisfaction outcomes for the whole held-out set and recompute the satisfaction rate within each level of `Class`, `Customer Type` and `Type of Travel`. This gives a posterior predictive distribution for each subgroup rate, against which the observed rate can be compared.

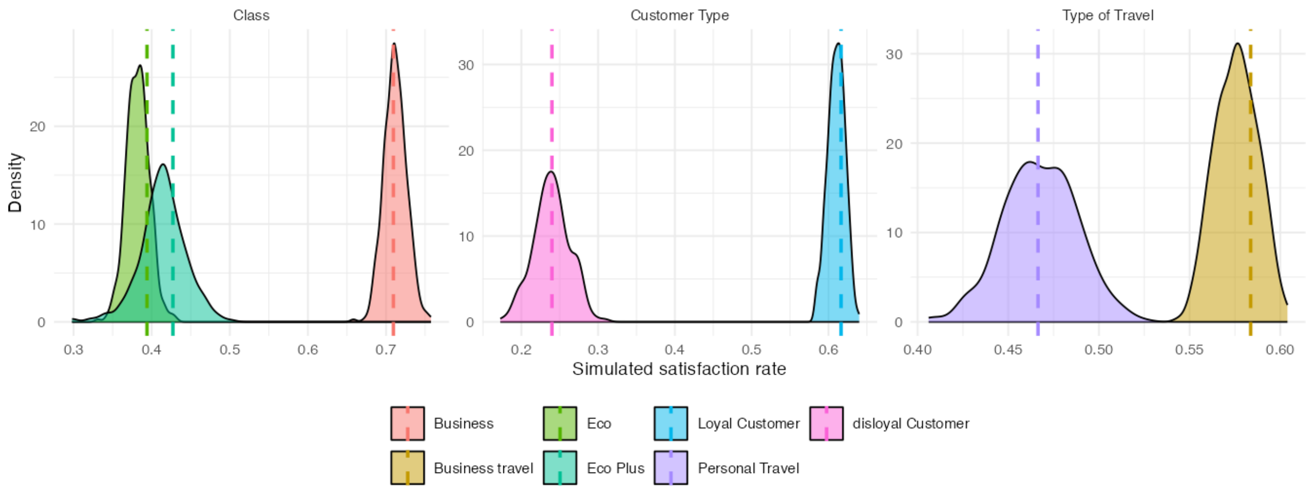


Figure 7. Posterior predictive check by subgroup. Densities are the simulated satisfaction rates; dashed lines mark the rates actually observed in the held-out set.

The observed rates fall inside the bulk of the corresponding simulated distributions for every subgroup of all three variables. The model is therefore not systematically over- or under-predicting satisfaction for any class of customer, on data it has never seen. This is a genuine adequacy check rather than a restatement of accuracy: the subgroup rates were never fitted directly, so reproducing them is evidence that the covariate effects have been captured correctly rather than merely tuned to the aggregate.

5.6 Choosing the cutoff by expected loss

Section 5 deferred the choice of operating point. A decision-theoretic treatment settles it properly: instead of picking 0.5 or the ROC-optimal point by convention, we state what the two errors cost and minimise posterior expected loss.

Suppose the airline flags a customer as at risk when $\hat{p}_i < t$ and acts on that flag with a retention contact. Write C_{miss} for the cost of failing to flag a customer who is in fact dissatisfied, a lost retention opportunity, and C_{waste} for the cost of contacting a customer who was satisfied anyway. The expected loss at cutoff t is

$$\mathscr{L}(t) = C_{\text{miss}} \cdot \mathbb{P}\big(\hat{p} \geq t,\ y=0\big) + C_{\text{waste}} \cdot \mathbb{P}\big(\hat{p} < t,\ y=1\big), \tag{8}$$

estimated on the held-out set. Only the *ratio* of the two costs matters for the location of the optimum, which is convenient: the airline need not put a currency figure on either, only state how many wasted contacts it would trade for one retained customer.

Under equal costs the optimum is $t^{\star} = 0.56$, essentially the base rate and close to the conventional 0.5, which is a useful sanity check on Equation (8). As missing a dissatisfied customer becomes more expensive the cutoff rises, to 0.74 at a 3 : 1 ratio and 0.80 at 5 : 1: the airline should flag far more customers, accepting many wasted contacts in exchange for catching the recoverable ones.

Table 8. Cutoff minimising expected loss, by assumed cost ratio $C_{\text{miss}} : C_{\text{waste}}$.

Cost ratio	Optimal cutoff $t^\star$	Expected loss
1 : 1 (equal cost)	0.56	0.172
3 : 1	0.74	0.252
5 : 1	0.80	0.287

This is where the Bayesian machinery pays off in operational terms. Because the model returns a calibrated distribution rather than a label, the same fit supports any cost structure the business cares to specify, and the cutoff follows from the loss function instead of being fixed by convention.

6. Conclusions

We modelled airline customer satisfaction with a Bernoulli-logit likelihood, first on the reduced covariate set suggested by the brief and then on the full set under a horseshoe shrinkage prior, with `BAS` model averaging as an independent cross-check. All posteriors were sampled by MCMC in JAGS.

Three effects dominate and survive every specification. Customer loyalty is the strongest by a wide margin and the least shrunk coefficient at every global scale tested; *Business* class follows; and among the fourteen service items *Inflight entertainment* is the one the prior preserves most clearly, while *Inflight wifi service* and *Gate location* are heavily shrunk despite sitting in the same correlated clusters. That asymmetry is what the horseshoe was chosen for.

These conclusions are not artefacts of our modelling choices. The vague-prior posterior moves by at most 0.066 on the log-odds scale across four orders of magnitude of prior variance; only one of twenty-two predictors changes its signal/noise verdict across a hundred-fold range of the horseshoe's global scale, and it sits on the $\kappa = 0.5$ boundary; and refitting on a training set twelve times larger moves the coefficients we report by at most 0.022 while narrowing the intervals by the factor $\sqrt{n}$ asymptotics predicts. Out of sample, on 128 274 unseen customers, the model reaches an AUC of 0.900 and classifies 82.8% correctly, and the posterior predictive checks reproduce the observed satisfaction rate in every subgroup.

One result resisted explanation. *Departure/Arrival time convenient* carries a negative coefficient whose interval excludes zero. We suspected the structural zeros and tested it by refitting without those rows; the effect strengthened rather than weakened, so we report it as an open question.

Three limitations are worth naming: the structural zeros were flagged and tested but not modelled with a zero-inflated likelihood; $n \approx 1200$ is modest for twenty-two predictors, which is why several effects are borderline; and the encoding and link choices of Section 2 were settled on training WAIC without out-of-sample re-validation.

A. Data augmentation

A.1 Decorrelating departure and arrival delays

Inspecting the correlation matrix (Fig. 2a), we notice an extremely high correlation between `Departure delay` and `Arrival delay`. This value makes the two variables almost duplicates of one another. In order to remove this redundancy while retaining as much information as possible, we decided to replace Arrival delay with a quantity that captures the difference between the two.

We considered two options:

- using the plain difference between the two variables;
- fitting a simple linear regression model with `Arrival delay` as the target and `Departure delay` as the covariate, and taking the residual of each observation as the new covariate.

The second model is more general and, in fact, encompasses the first one; however, the first is more interpretable and carries an immediate meaning, such as “mid-flight delay” or “time gained during the flight”. The deciding point was whether the second option differed enough from the first to justify its higher complexity.

The regression model that reproduces the plain difference is the one with slope equal to 1 and intercept equal to 0. Fitting the regression model on the full dataset yields an intercept of 0.76 and a slope of 0.97; we therefore decided to opt for the simple difference between the arrival and departure delays.

A.2 Reducing covariates skeweness

From the boxplots in Fig.8 we can see how covariates `Flight Distance`, `Departure Delay` and `Mid-flight delay` are extremely right-skewed. Since covariate skewnes might slow down MCMC convergence and are prone to present outliers and leverage points, we decided to transform them and check whether the transformation is beneficial.

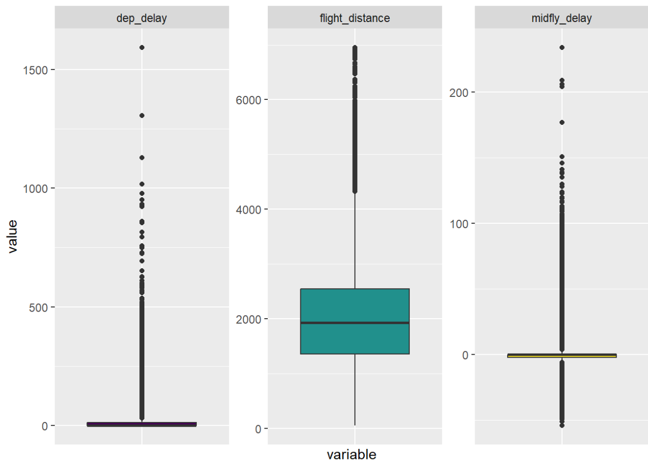


Figure 8. Boxplots of skewed covriates.

Since both `Flight distance` and `Departure delay` are non-negative we transform them using $ln(1+x)$; instead for `Mid-flight delay` we use the *signed-log transformation* $\text{sign}(x) \cdot ln(1+|x|)$ due the presence of negative values.

variable	skewness
Flight Distance	0.466
log1p(Flight Distance)	-1.599
Departure Delay	6.853
log1p(Departure Delay)	0.919
Midflight Delay	2.941
signed-log(Midflight delay)	0.234

Table 9. Skewness of raw and transformed variables

As we can see in Tab.9 `Flight distance` is only midly skewed and the transformation actually turn to be worse than the original varibale, therefore we kept it unchanged. Instead for what concerns `Departure delay` and `Mid-flight delay` we see higher level of skeweness and the transformation improves this aspect; therefore we decided to kee the log-transformed variables.

B. Feature selection cross-check with BAS

After fitting our feature-selection model based on a horseshoe prior, we would like to compare it with a Bayesian Adaptive Sampling (BAS) model in order to verify whether the two methods agree on which covariates are more useful in explaining the target variable.

With 23 predictors, $2^{23} \approx 8.4$ million candidate models are far too many to enumerate exhaustively, so we use `method = "MCMC"` in BAS to stochastically search the model space. A later section assesses whether this search suffers from instability or not.

Table 10 reports the marginal posterior inclusion probabilities $P(\beta_j \neq 0 \mid Y)$ for every predictor, together with the five highest-probability models returned by the search and their summary statistics (Bayes factor, posterior probability, R^2, dimension and log marginal likelihood).

Table 10. Marginal posterior inclusion probabilities and the top five models identified by the BAS MCMC search. A value of 1.000/0.000 in a model column indicates the predictor is included/excluded in that model.

	$P(\beta \neq 0 \mid Y)$	Model 1	Model 2	Model 3	Model 4	Model 5
Intercept	1.000	1.000	1.000	1.000	1.000	1.000
seat_comfort	0.803	1.000	1.000	1.000	1.000	0.000
time_convenient	0.973	1.000	1.000	1.000	1.000	1.000
food_drink	0.500	1.000	0.000	1.000	0.000	0.000
gate_location	0.037	0.000	0.000	0.000	0.000	0.000
wifi	0.049	0.000	0.000	0.000	0.000	0.000
entertainment	1.000	1.000	1.000	1.000	1.000	1.000
online_support	0.220	0.000	0.000	0.000	0.000	0.000
ease_booking	0.954	1.000	1.000	1.000	1.000	1.000
onboard_service	0.996	1.000	1.000	1.000	1.000	1.000
legroom	0.965	1.000	1.000	1.000	1.000	1.000
baggage	0.073	0.000	0.000	0.000	0.000	0.000
checkin	0.986	1.000	1.000	1.000	1.000	1.000
cleanliness	0.310	0.000	0.000	0.000	0.000	0.000
online_boarding	0.029	0.000	0.000	0.000	0.000	0.000
Age_std	0.037	0.000	0.000	0.000	0.000	0.000
FlightDist_std	0.035	0.000	0.000	0.000	0.000	0.000
Class_EcoPlus	0.029	0.000	0.000	0.000	0.000	0.000
Class_Business	0.993	1.000	1.000	1.000	1.000	1.000
logdep_std	0.092	0.000	0.000	0.000	0.000	0.000
TypeTravel_Personal	0.451	0.000	0.000	1.000	1.000	1.000
CustType_disloyal	1.000	1.000	1.000	1.000	1.000	1.000
logmid_delay	0.031	0.000	0.000	0.000	0.000	0.000
BF	NA	1.000	0.574	0.477	0.369	0.268
PostProbs	NA	0.128	0.071	0.059	0.048	0.033
R^2	NA	0.426	0.421	0.430	0.425	0.420
dim	NA	11.000	10.000	12.000	11.000	10.000
logmarg	NA	−516.838	−517.392	−517.578	−517.835	−518.155

C. Parameter-space convergence

Table 11. MCMC settings used for the logit model

Setting	Value
Chains	3
Adaptation	1500
Burn-in	1000
Iterations (kept)	3000
Thinning	7

We now test the goodness of the MCMC search over the parameter space executed by BAS.

Convergence within one run

BAS reports two independent estimates of each posterior inclusion probability:

- `probne0.RB`: takes the set of distinct visited models that contain variable j, uses their marginal likelihoods, renormalises and sums the mass of those models;
- `probne0.MCMC`: the raw fraction of iterations the chain spent in some model containing variable j.

Table 12. MCMC settings used for the horseshoe model

Setting	Value
Chains	3
Adaptation	1500
Burn-in	1000
Iterations (kept)	5000
Thinning	3

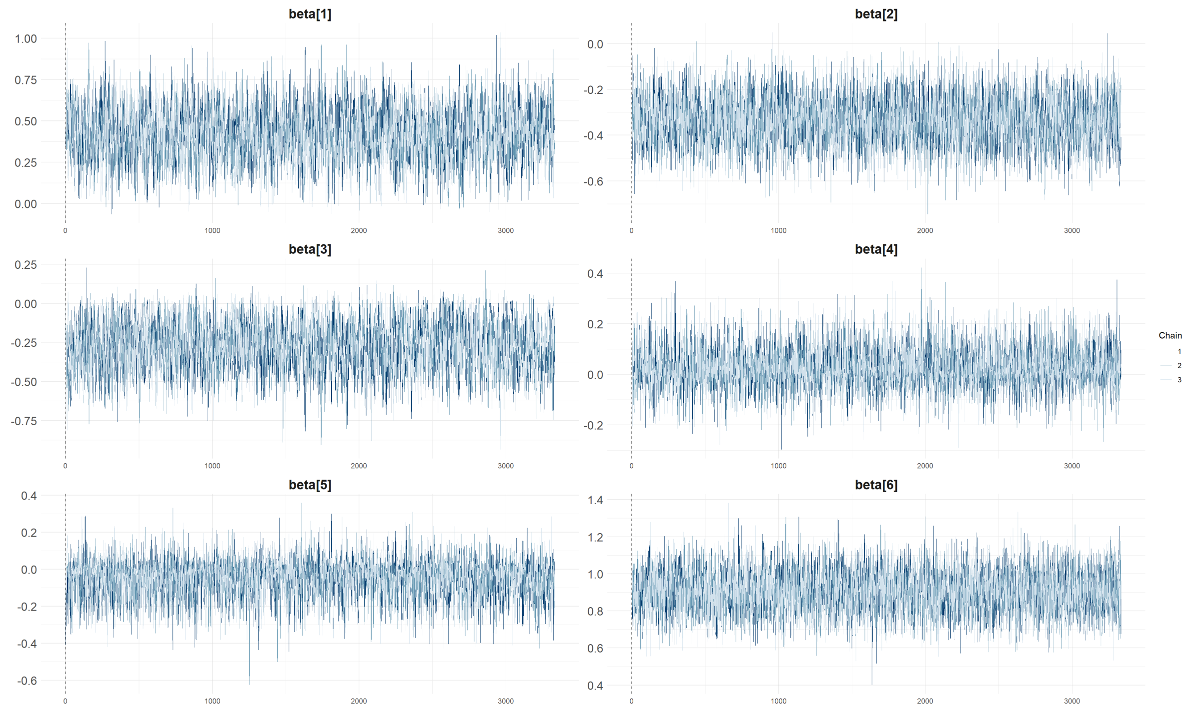


Figure 9. Representative trace plot for a subset of the regression coefficients β_j under the horseshoe prior (three chains).

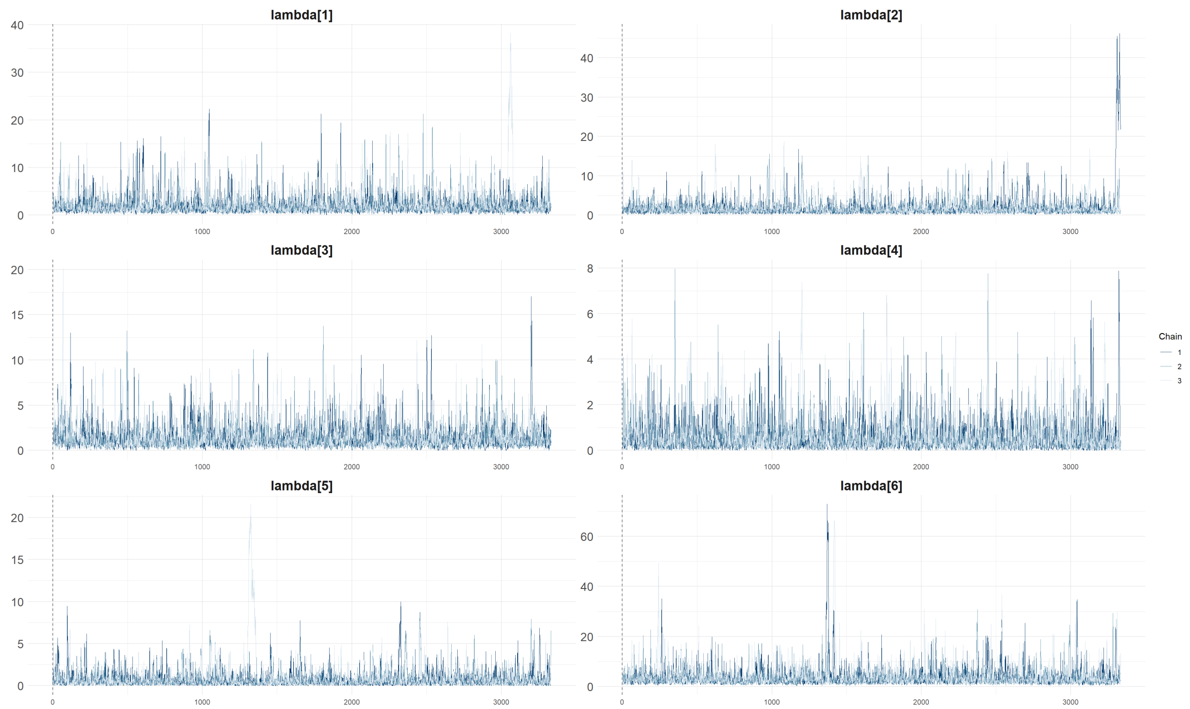


Figure 10. Representative trace plot for the local scales λ_j and the global scale τ (three chains). These scale parameters mix more slowly than the coefficients.

Table 13. Posterior shrinkage weight κ_j at each global scale (one row per predictor; 22 predictors in total).

Term	$s = 0.1$	$s = 1$	$s = 10$
disloyal Customer	0.052	0.071	0.076
Inflight entertainment	0.133	0.171	0.181
Class: Business	0.145	0.180	0.191
On-board service	0.312	0.364	0.378
Personal Travel	0.316	0.349	0.362
Seat comfort	0.326	0.365	0.382
Ease of Online booking	0.361	0.416	0.429
Checkin service	0.370	0.426	0.434
Dep/Arr time convenient	0.371	0.418	0.419
Leg room service	0.397	0.449	0.465
Food and drink	0.449	0.489	0.489
Online support	0.485	0.525	0.534
Cleanliness	0.509	0.544	0.550
Class: Eco Plus	0.653	0.667	0.667
Baggage handling	0.673	0.692	0.703
log(1+Departure Delay)	0.684	0.700	0.717
Inflight wifi service	0.705	0.740	0.739
Online boarding	0.714	0.728	0.732
Age	0.735	0.765	0.766
Flight Distance	0.751	0.752	0.770
Gate location	0.751	0.756	0.750
Mid-flight delay (signed log)	0.770	0.780	0.783

If the chain has converged, the two estimates should agree closely.

The correlation between the two estimates is 0.9999 and the maximum absolute difference is 0.014: the points sit essentially on the diagonal (Figure 11), the sign that this run has converged.

Stability across seeds

We refit the identical model twice more with different random seeds and compare the posterior inclusion probabilities (PIPs) to check whether they agree. Table 14 lists the predictors with the largest spread, ordered by the range across the three seeds.

Table 14. Posterior inclusion probabilities across three random seeds, with the range (max − min) per predictor. Only the eight most variable predictors are shown.

term	seed123	seed1	seed2	range
seat_comfort	0.803	0.841	0.826	0.038
food_drink	0.500	0.533	0.511	0.033
online_support	0.220	0.228	0.204	0.024
cleanliness	0.310	0.297	0.316	0.019
TypeTravel_Personal	0.451	0.439	0.457	0.017
legroom	0.965	0.951	0.959	0.013
time_convenient	0.973	0.979	0.983	0.011
baggage	0.073	0.065	0.064	0.009

BIC prior as a robustness check

We try another prior on the β coefficients: if the search is stable, the effect of the prior should vanish once the number of iterations is large enough. Table 15 compares the g-prior PIPs with those obtained under a BIC-based prior.

With 100,000 MCMC iterations, PIPs move by at most 0.038 across the three seeds and by at most 0.004 between the g-prior and the BIC-based prior. Given the number of MCMC iterations, the search proves itself to be stable.

D. BAS vs. Horseshoe: model comparison

Finally, we compare the models selected by the two approaches and check on which coefficients they agree and on which they do not. Figure 12 plots each predictor's BAS inclusion probability against the horseshoe's $1 - \kappa_j$ shrinkage measure, colouring predictors by whether the two methods make the same signal/noise call.

91% of the 22 predictors get the same signal/noise call from BAS's model-space search and from the horseshoe's continuous shrinkage - two structurally different estimation strategies (discrete model averaging vs. a single shrinkage prior) landing

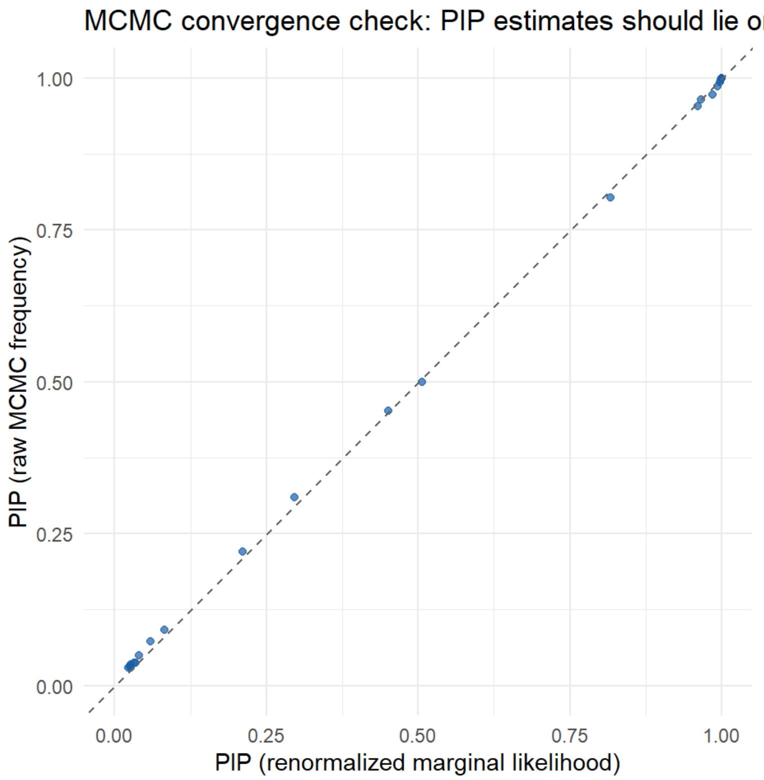


Figure 11. MCMC convergence check. Renormalised-marginal-likelihood inclusion probabilities (x-axis) against raw MCMC-frequency inclusion probabilities (y-axis); points lying on the dashed diagonal indicate the two estimates agree.

Table 15. Posterior inclusion probabilities under the g-prior and the BIC-based prior, with the absolute difference per predictor (six most affected predictors shown).

term	g_prior	BIC	abs_diff
online_support	0.220	0.224	0.004
cleanliness	0.310	0.314	0.004
seat_comfort	0.803	0.801	0.002
food_drink	0.500	0.498	0.002
ease_booking	0.954	0.952	0.002
legroom	0.965	0.967	0.002

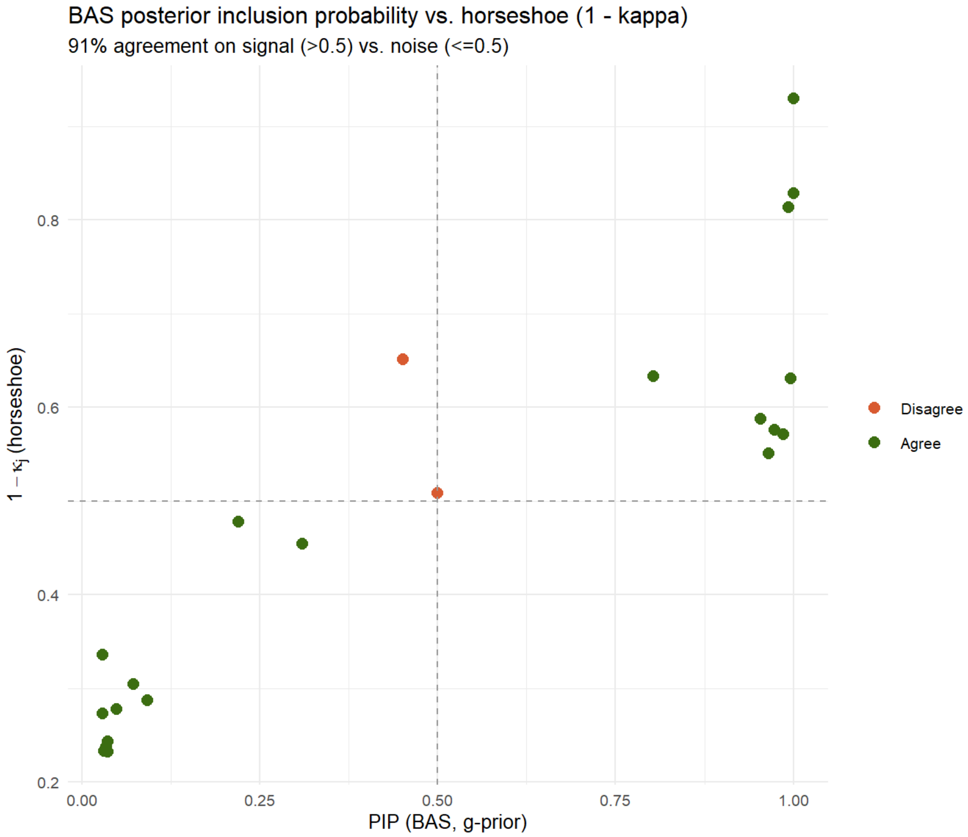


Figure 12. BAS posterior inclusion probability (x-axis) against the horseshoe's $1-\kappa_j$ (y-axis). Dashed lines mark the 0.5 signal/noise cutoff for each method; green points denote agreement, the red point denotes the single disagreement.

on the same handful of variables. The one disagreement, `Class: Business` (PIP 0.993, $\kappa = 0.186$), is a genuine borderline case in both methods (BAS just below its 0.5 cutoff, the horseshoe's credible interval for it crossing zero in Section 03) - not a real contradiction between them.

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[1] M. Villan, *Bayesian Learning Book*. [Online]. Available: https://mattiasvillani.com/BayesianLearningBook/.

[2] *Airline customer satisfaction*. [Online]. Available: https://www.kaggle.com/datasets/raminhuseyn/airline-customer-satisfaction.